

Kaartiigeyan S V

VLSI Engineer

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SUMMARY

As an Analog IC Design Engineer, I excel in developing high-performance semiconductor solutions, leveraging my deep understanding of CMOS technology and RTL design methodologies. My goal is to integrate analog circuits with digital electronics to deliver top-notch products.

SKILLS

Programming Languages	Python, System Verilog, Verilog, C, TCL, Bash, Perl
Design Methodologies	RTL Design, CMOS Technology, Analog Circuit Design, Digital Electronics, FPGA Validation
Soft Skills	Analytical Skills, Problem-Solving, Teamwork, Leadership
Languages	English, Kannada, Hindi, Tamil, Telugu

EDUCATION

B.Tech (ECE)

NMAM Institute of Technology, Udupi 2023 - Ongoing

- Currently pursuing a **Bachelor's degree in Electronics and Communication Engineering** (B.Tech - ECE) at NMAM Institute of Technology, Udupi, achieving a **CGPA of 7.81**. Focused on foundational aspects of **electronic systems and communication technologies**.

12th Grade

Christ Academy Junior College 2021 - 2023

10th Grade

St. Francis De Sales Public School 2009 - 2021

PROJECTS

1.UVM Testbench for PicoRV32

Ongoing

- Currently engaged in developing a **UVM testbench** as part of my final year major project, focusing on the **functional verification** of a **RISC-V processor core**.
- ### 2.Bandgap Reference IC Design & Layout (SKY130, 130nm)
- SKY130, 130nm, DRC, LVS, Xschem, Magic, ngspice
- Designed and laid out a cascode-compensated bandgap voltage reference (avbdbgp_3v3_sky130_v1) in SKY130 130nm CMOS**, using Xschem for schematic capture and Magic for full-custom layout, achieving PVT-stable output via PTAT/CTAT current cancellation.
 - Implemented common-centroid layout of matched BJT devices** (unit-area core transistors split and interdigitated) to minimize gradient-induced mismatch and improve PSRR and temperature stability of the reference voltage.
 - Added cascode current mirrors to the core PTAT/CTAT generation loop**, improving output impedance and current-mirror matching accuracy, directly reducing systematic error in the bandgap's temperature coefficient.
 - Performed DRC/LVS closure and post-layout parasitic extraction (PEX)** in Magic, generating SPICE netlists with distributed R/C parasitics and re-simulating loop stability (phase margin) and PSRR in ngspice to validate compensation robustness against layout-induced parasitics.
 - Verified circuit-to-layout equivalence and analyzed compensation trade-offs** (Miller/cascode compensation) under real parasitic loading, closing the loop between analog design intent and physical implementation — demonstrating end-to-end analog IC layout flow from schematic to sign-off.

3.Router 1x3 Design

- Architected a 1x3 packet router in Verilog HDL** with a modular 6-block design (3 FIFOs, synchronizer, control register, FSM), handling packet decoding, routing, reset, and read/write control paths independently for scalability and reuse.
- Designed asynchronous FIFOs with a dedicated synchronizer block** to safely handle clock domain crossing between packet-receive and packet-dispatch logic, preventing metastability in cross-domain signal transfer.

- **Implemented a Finite State Machine (FSM) as the central control unit**, sequencing packet validation, routing decisions, and reset behavior — demonstrating strong grasp of RTL control-path design and state-based protocol handling.
- **Built and verified the design using [Vivado]** with functional simulation testbenches covering packet transmission, corruption/error checks, and reset scenarios, ensuring correct routing behavior across all 3 output channels.
- **Applied digital design fundamentals (FIFO buffering, register-based storage, synchronization, FSM control)** to a real-world packet-switching problem, bridging RTL design skills with system-level architecture thinking relevant to SoC/NoC interconnect design.

WORK EXPERIENCE

Machine Learning for Advance Process Control(MLAPC) Lab,MIT

Research Intern Nov 2025 - Present

- Engineering a Control-Informed Reinforcement Learning (CIRL) agent by embedding a Proportional-Integral-Derivative (PID) controller layer directly into a deep neural network architecture.
- Automating dynamic gain scheduling by training the neural network to output continuous adjustments for PID parameters in real-time.
- Implementing a derivative-free policy optimization algorithm that combines random search and Particle Swarm Optimization (PSO) to train the neural network without requiring backpropagation through the PID layer.
- Designing a custom reward function to optimize the trade-off between set point tracking accuracy and overall control effort.

CERTIFICATIONS

System Verilog for Verification

2026 - Ongoing

ACHIEVEMENTS

University Powerlifter

Technical Head(AY 2026-27)-Electronics and Communication Students Association

Joint Treasurer(AY 2024-25)-Electronics and Communication Students Association

Member of college basketball team